BT Practical 19: Survey Report on Types of Blockchains and their Real-Time Use Cases

Type: No code. This is a written report (usually 4–6 handwritten or typed pages).

Steps to do

1. **Research:** Read about the 4 types of blockchain (public, private, consortium, hybrid) from your textbook or class notes,

https://ethereum.org/en/developers/docs/, https://www.hyperledger.org and IBM's blockchain pages.

2. **Collect use cases:** Find 2 real projects for each type (examples are in the table below).
3. **Write the report** using the structure below. Write in your own words.
4. **Draw 2 diagrams:** (a) the structure of a block chain (blocks linked by hashes), (b) a comparison chart of the 4 types.
5. **Add the comparison table** and a conclusion. List your references at the end.

Report structure (headings to use)

1. **Title page:** title, name, roll no, date
2. **Introduction:** what a blockchain is: a distributed, immutable ledger; blocks linked by hashes; consensus
3. **Structure of a block:** block header (previous hash, timestamp, nonce, Merkle root) plus a list of transactions
4. **Types of blockchain:** one paragraph each:
   - **Public (permissionless):** anyone can read, write and validate. *Bitcoin, Ethereum.* Pros: fully decentralized, transparent. Cons: slow, high energy use (PoW).
   - **Private (permissioned):** one organization controls who joins. *Hyperledger Fabric, Ripple (partly).* Pros: fast, private. Cons: centralized.
   - **Consortium (federated):** a group of organizations controls it. *R3 Corda, Energy Web Chain, banking consortia.* Pros: shared trust, efficient. Cons: needs agreement between members.
   - **Hybrid:** private data with public verification. *Dragonchain, XinFin (XDC).* Pros: flexible. Cons: complex.
5. **Consensus mechanisms:** PoW, PoS, PBFT, PoA, each in 2 lines
6. **Comparison table** (copy the table below)
7. **Real-time use cases** (table below, plus 2–3 lines each)
8. **Advantages and limitations of blockchain**
9. **Conclusion**
10. **References**

Comparison table

Feature	Public	Private	Consortium	Hybrid
Who can join	Anyone	Invited only	Member organizations	Mixed
Controlled by	No one (community)	Single organization	Group of organizations	Organization + public layer
Speed	Slow	Fast	Fast	Medium–Fast
Decentralization	High	Low	Medium	Medium

Feature	Public	Private	Consortium	Hybrid
Consensus	PoW / PoS	PBFT / Raft / PoA	PBFT / voting	Combination
Data privacy	Fully transparent	Private	Shared among members	Selective
Examples	Bitcoin, Ethereum	Hyperledger Fabric	R3 Corda	Dragonchain, XinFin

Real-time use cases

Domain	Use case	Type used	Example
Finance	Cross-border payments	Public / Consortium	RippleNet, JPMorgan Onyx
Cryptocurrency	Digital money	Public	Bitcoin, Ethereum
Supply chain	Tracking food from farm to shop	Private / Consortium	IBM Food Trust (Walmart)
Healthcare	Secure patient records	Private / Hybrid	MedRec, BurstIQ
Government	Land records, e-voting	Private / Hybrid	Andhra Pradesh land registry, Estonia e-Residency
Identity	Self-sovereign digital ID	Public / Hybrid	Microsoft ION
Energy	Peer-to-peer energy trading	Consortium	Energy Web Chain, Power Ledger
NFTs / Gaming	Ownership of digital assets	Public	OpenSea, Axie Infinity

Conclusion (write this)

Blockchains are of four types: public, private, consortium and hybrid. They differ in who can join, who controls the network and how fast and private it is. Public chains suit open cryptocurrencies. Private and consortium chains suit businesses that need speed and privacy. Hybrid chains combine both. Blockchain is now used in finance, supply chain, healthcare, government and identity management.